Unit 1 – Using Objects and Methods
Lesson 1: Welcome to Java



Code	**Console**

1: public class Example {
2: public static void main(String[] args) {
3: //displays to the screen
4: System.out.println("Welcome to AP CSA!");
5: }
6: }
	Welcome to AP CSA!

- Line 1 defines a _____.
- Every Java program must have _____ class.
- Each class has a name and it should start with an uppercase letter.
- Our classes will always start with _____.
- In our program, the class name is _____.

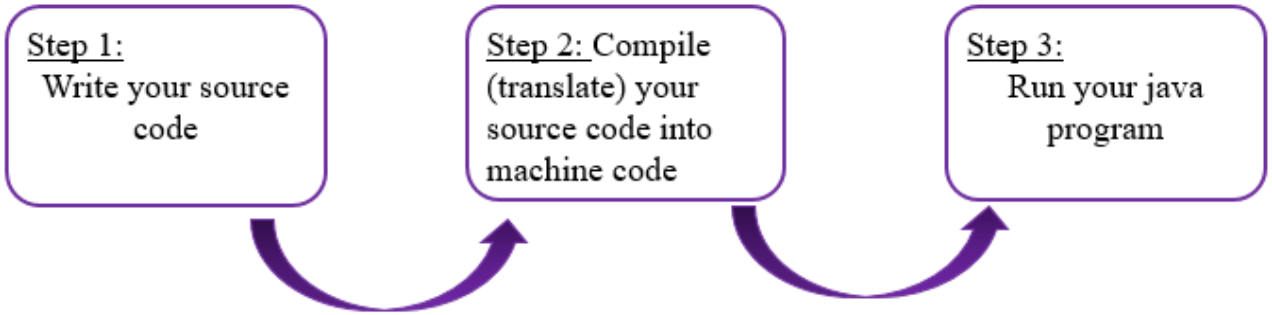
- Line 2 defines the _____.
- All Java programs execute from the main method ONLY.
- There is _____ main method per program.
- These keywords should always be part of your main method signature.

- Line 3 is a _____.
- Comments help programmers communicate and understand their program.
- A comment that is preceded by two slashes (//) is called a _____.
- A comment that is enclosed between /* and */ and can span several lines is called a _____.
- Comments are not required in a program.

- Line 4 is a _____.
- This statement uses the code `System.out.println`() to display a _____.
- A _____ is a sequence of characters enclosed in quotations.
- Every statement in Java ends in a _____.

Special Characters

Character	*Name*	*Description*	
	Open and close braces	• These curly braces form a block. • Each block begins with an open brace and ends with a close brace. • In the example above, there is a class block and a method block.	• Java has its own _____ • You have to write code that conforms to special characters and formats • If you violate a syntax rule, you will get a _____ • A compiler error means that the compiler cannot translate your code into _____
	Open and close brackets	• These create an array	
	Open and close parentheses	• These have multiple uses in a program, but above they are used with our methods.	
	Double slash	• Denotes a line comment. • Everything on the line after the double slashes will be ignored by the program and will only be able to be seen by the programmer.	
	Open and close quotations	• These enclose a string literal	
	Semicolon	• Marks the end of a statement	



Types of Errors

Error	*Definition*	*Examples*	*Solution*
	a mistake in the program where the rules of the programming language are not followed	• Missing semicolons • Missing brackets • Misspelled words	The compiler will catch these before the program runs
	a mistake in program that causes it to behave incorrectly or unexpectedly	• Flawed reasoning in the code • Misplaced loop boundary • Off by one error in arrays	Test the program with different types of data to make sure
	a mistake in the program that occurs during the execution of a program	• Infinite loop • Stack Overflow • Out Of Memory	These errors will cause the program to terminate abnormally
	a type of run-time error that occurs as a result of an unexpected error, but can be predicted and handled	• ArithmeticException • NullPointerException • IOException	Using a try/catch block, you can handle exceptions if they come up in your program

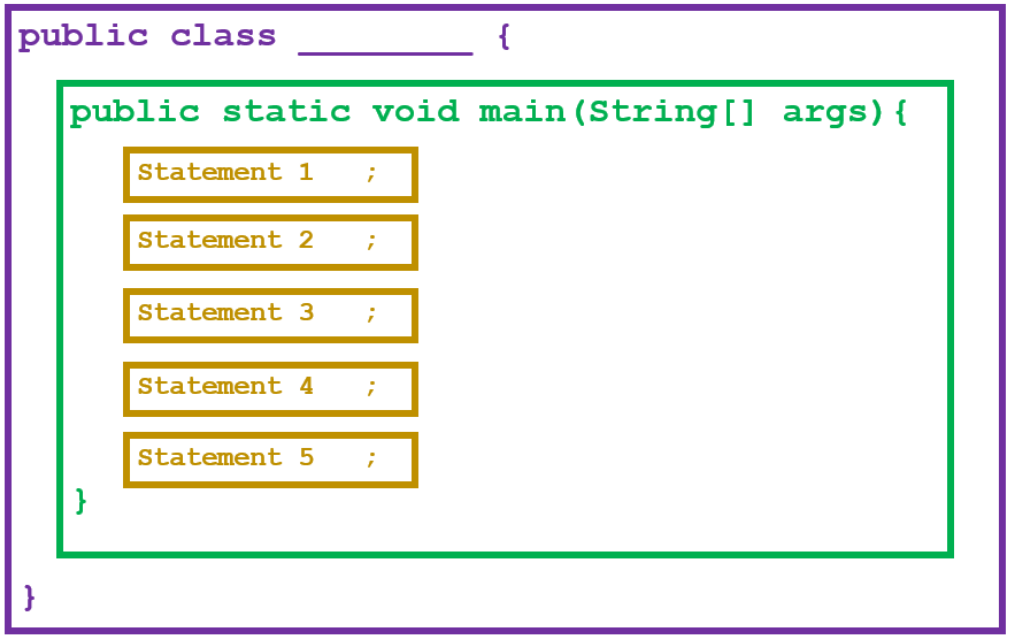
Styling

- Good style makes is easy for you and others to read
- Align the blocks of code, along with the brackets for the class and methods, for easy readability
- Java ignores whitespace, so tabs and newlines are purely for readability

For example, our first program could have looked like this:

```
public class Example{public static void main(String[] args){System.out.println("Welcome to AP CSA!");}}
```

The above is difficult to read and the block format we created was much better:



There are two popular styles when it comes to blocking, and each is considered good programming style; it is really up to your personal preference.

Next-Line Style	**End-of-line Style**

public class Example
{
public static void main(String[] args)
{
//displays to the screen
System.out.println(“Welcome to AP CSA!”);
}
}
	public class Example {
	public static void main(String[] args) {
	//displays to the screen
	System.out.println(“Welcome to AP CSA!”);
	}
	}